

Booking_receipt
booking_id
passanger_full_name
passanger_passport
flight_number
departure_airport_name
departure_city
arrival_airport_name
arrival_city
airline_name
seat_numbers
ticket_price

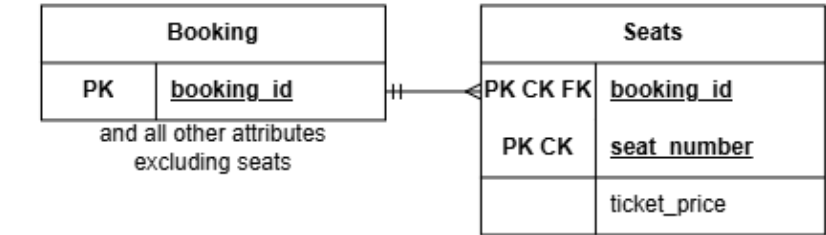
booking_id	passanger_fullname	passanger_passport	flight_number	dep_airport_name	dep_city	arr_airport_city	arr_city	airline_name	seat_numbers	ticket_price
1	Lady Bug	023	2	CDG	Paris	PEK	Peking	Air1	C12,C15	500
2	Cat Noir	456	2	CDG	Paris	PEK	Peking	Air1	C13	450
3	Rena Rouge	432	5	ORY	Paris	ALA	Almaty	Air3	B16	1200
4	Queen Bee	753	2	CDG	Paris	PEK	Peking	Air1	B4	450
5	King Monkey	159	4	NCE	Nice	MLA	Milan	Air9	E8,E9	300
6	Miss Hound	789	2	CDG	Paris	PEK	Peking	Air1	A5	450

booking_id	passanger_fullname	passanger_passport	flight_number	dep_airport_name	dep_city	arr_airport_city	arr_city	airline_name	seat_number	ticket_price
1	Lady Bug	023	2	CDG	Paris	PEK	Peking	Air1	C12	250
1	Lady Bug	023	2	CDG	Paris	PEK	Peking	Air1	C15	250
2	Cat Noir	456	2	CDG	Paris	PEK	Peking	Air1	C13	450
3	Rena Rouge	432	5	ORY	Paris	ALA	Almaty	Air3	B16	1 200
4	Queen Bee	753	2	CDG	Paris	PEK	Peking	Air1	B4	450
5	King Monkey	159	4	NCE	Nice	MLA	Milan	Air9	E8	150
5	King Monkey	159	4	NCE	Nice	MLA	Milan	Air9	E9	150
6	Miss Hound	789	2	CDG	Paris	PEK	Peking	Air1	A5	450

1. table violates 1NF Reason: It has seat_numbers column that stores comma-separeted list with several values in one cell. When 1NF have to be atomic and single-valued

violates 2NF Reason: Partian dependencies. Non-key attributes depend obly on part of a composite primary key.
Example: flight number, dep airport name, der city, arr airport name, arr city and airline name depend only on booking but not also passangers.

violates 3NF Reason: Transitive dependencies: Non-key attributes depend on other non-key attributes. Such as dep city rely on dep airport, and same situation with arrivals.



3. Functional Dependencies (FDs):

1NF to 2NF:
booking_id -> passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name
(booking_id, seat_number) -> ticket_price

2NF to 3NF:
passenger_passport_number -> full_name
departure_airport_name -> departure_city
arrival_airport_name -> arrival_city
flight_number -> airline_name, departure_airport_name, arrival_airport_name
booking_id -> passenger_passport_number, flight_number
(booking_id, seat_number) -> ticket_price



4. 3NF Analysis for Flights:

PK: flight_number
FDs: flight_number to Airline_name, dep_airport, arr_airport

Conclusion: The Flights table is in 3NF.

Explanation: All non-key attributes are fully functionally dependent directly on the PK. There are no transitive dependencies between non-key attributes. The city attributes dep_city, arr_city that depended on airport codes were moved to the Airports table. So, dep_airport and arr_airport act strictly as FKs to the Airports table without introducing transitive dependencies in Flights.